

Chain of Thought (CoT) and Tree of Thought (ToT) Prompting Worksheet

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Part A: Chain of Thought (CoT) Prompting

Rewrite each basic prompt into a Chain of Thought prompt that asks the AI to reason step-by-step.

CoT Task 1

Basic Prompt: A train travels 120 km in 2 hours. What is its average speed?

Rewrite: Calculate the average speed step by step. First identify the given distance and time, then apply the average speed formula ($\text{Speed} = \text{Distance} \div \text{Time}$), perform the calculation, and finally provide the answer with the correct answer.

CoT Task 2

Basic Prompt: I earn ₹50,000 per month. My expenses are ₹30,000. How much can I save in one year?

Rewrite: Solve this step by step. First calculate my monthly savings by subtracting expenses from income. Then multiply the monthly savings by 12 to find the total annual savings. Show each calculation clearly before giving the final answer.

CoT Task 3

Basic Prompt: A company sells a product for ₹500. Cost price is ₹350. Calculate profit percentage.

Rewrite: Through the calculation step by step. First find out the profit by subtracting the cost price from the selling price. Then apply the profit percentage formula ($\text{Profit} \div \text{Cost Price} \times 100$), perform the calculation, and present the final profit percentage.

CoT Task 4

Basic Prompt: Should I buy a laptop or a tablet for college?

Rewrite: Through the decision step by step. Compare a laptop and a tablet based on academic work, note-taking, programming, specifications, battery life, cost, and long-term usefulness for college. Explain the advantages and disadvantages of each option before giving a recommendation.

CoT Task 5

Basic Prompt: Which career should I choose after B.Tech?

Rewrite: Through the decision step by step. Consider factors such as my interests, skills, career goals, higher education opportunities, salary potential, job demand, and work-life balance. Compare the most suitable career options after B.Tech, explain the strengths and limitations of each, and then recommend the best choice with a clear identification.

Part B: Tree of Thought (ToT) Prompting

Rewrite each prompt so the AI explores multiple options, compares them, and recommends the best one.

ToT Task 1

Basic Prompt: Plan a college tech fest.

Rewrite: Explore multiple ways to organize a college tech fest. Consider different themes, event formats, budgets, sponsorship ideas, and promotional strategies. Compare the advantages and disadvantages of each approach, then recommend the best plan with a clear schedule and justification.

ToT Task 2

Basic Prompt: Suggest a startup idea.

Rewrite: Generate at least three startup ideas in different industries. For each idea, evaluate the target market, required investment, scalability, competition, and potential risks. Compare the ideas and recommend the most promising startup with reasons.

ToT Task 3

Basic Prompt: How should I prepare for exams?

Rewrite: Explore several exam preparation strategies, such as a daily study schedule, active recall, spaced repetition, group study, and practice tests. Compare their strengths and weaknesses based on effectiveness and time requirements, then recommend the best strategy or combination of strategies.

ToT Task 4

Basic Prompt: Create a marketing strategy for a new smartphone.

Rewrite: Develop multiple marketing strategies for launching a new smartphone, such as social media campaigns, influencer marketing, online advertising, and offline promotions. Compare each strategy based on cost, audience reach, expected impact, and return on investment, then recommend the most effective marketing plan.

ToT Task 5

Basic Prompt: Which AI tool should I use?

Rewrite: Consider multiple AI tools for my needs, such as ChatGPT, Google Gemini, Claude, Microsoft Copilot, and Perplexity AI. Compare them based on features, ease of use, accuracy, pricing, coding support, research capabilities, and content generation. Explain the strengths and limitations of each tool, then recommend the best option for my specific use case.